

Q1. Solve the system

$$\begin{pmatrix} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 2 & 1 & -1 & 8 \\ 0 & 0.5 & 0.5 & 1 \\ 0 & 2 & 1 & 5 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 2 & 1 & -1 & 8 \\ 0 & 0.5 & 0.5 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

~~back-sub: $z = 1$~~

back-sub: $z = -1$, $0.5y - 0.5 = 1 \Rightarrow y = 3$, $2x + 3 + 1 = 8 \Rightarrow x = 2$
 $x = 2$, $y = 3$, $z = -1$

Q2. Rank and nullity

rref has 2 pivots, so $\text{rank } A = 2$

$\text{nullity} = 3 - 2 = 1$

null space: $(-1, 2, 1)$

Q3. Subspace proof

$$W = \{ (x, y, z) : x + 2y - z = 0 \}$$

Need: closure under $+$ and under scalar multiplication.

Take $(1, 0, 1)$ and $(0, 1, 2)$, both in W .

Sum is $(1, 1, 3)$: $1 + 2 - 3 = 0$, so in W .

$3 \cdot (1, 0, 1) = (3, 0, 3)$: $3 + 0 - 3 = 0$, in W .

So W is a subspace.

Q4. Inverse

$$[A|I] =$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$